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Chapter 11 — Human Circulatory System

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter follows the textbook topics 11.1 to 11.5. Each topic is converted into board-style long questions, and each topic ends with conceptual short questions. Direct definitions are omitted.

Board pattern (Punjab 2026, new syllabus): Chapter 11 carries 2 MCQs, 3 short questions in Q.2, and part (b) of a long question in Q.5. Shorts are 2 marks; longs are 8 marks.

TOPIC 11.1 — STRUCTURE AND FUNCTIONING OF HEART

Textbook topic 11.1 · printed pages 231–238

Q.1 Describe the structure of the human heart, including pericardium, heart wall, chambers and valves.

Board long question · 8 marks

The heart is situated in the middle of the chest cavity between the lungs. It is enclosed in a sac called the **pericardium**, which separates it from surrounding organs.

Pericardium

- **Fibrous pericardium:** Outer strong connective tissue layer; protects against pressure and shocks and prevents excessive dilation.
- **Serous pericardium:** Inner sac with parietal and visceral layers. The visceral layer (epicardium) is attached to the heart. The **pericardial cavity** between them contains up to 50 mL of fluid that lubricates and protects the heart.

Heart wall

The wall has three layers: **epicardium** (outer), **myocardium** (thick cardiac muscle layer) and **endocardium** (inner epithelial lining).

Chambers and valves

The heart has two upper thin-walled **atria** and two lower thick-walled **ventricles**. The **tricuspid valve** lies between right atrium and right ventricle; the **bicuspid (mitral) valve** lies between left atrium and left ventricle. The **pulmonary** and **aortic semilunar valves** lie at the exits of the ventricles. These valves prevent backflow of blood.

The left ventricle wall is about three times thicker than the right because it pumps blood to the whole body, while the right ventricle pumps only to the lungs.

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Q.2 Explain the cardiac cycle, sounds of the heartbeat and the control of heartbeat.

Board long question · 8 marks

Cardiac cycle

The cardiac cycle consists of alternating relaxation and contraction. **Diastole** is the relaxed period when chambers fill with blood (about 0.4 s). **Atrial systole** (about 0.1 s) pushes blood into ventricles. **Ventricular systole** (about 0.3 s) pumps blood into pulmonary arteries and aorta.

Heart sounds

The **"lubb"** sound is produced by closure of tricuspid and bicuspid valves at the start of ventricular systole. The **"dubb"** sound is produced by closure of pulmonary and aortic semilunar valves at the end of ventricular systole. Incomplete valve closure may cause a **heart murmur**.

Control of heartbeat

The **SA node** (pacemaker) in the right atrium initiates impulses, causing atrial contraction. Impulses reach the **AV node**, then the **bundle of His** and **Purkinje fibres**, causing ventricular contraction. A delay of about 0.15 s at the AV node allows atrial systole to complete before ventricular systole. If the SA node is weak, an **artificial pacemaker** may be used.

An **ECG** records the electrical potentials generated by cardiac impulses and helps diagnose heart problems.

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Important Conceptual Short Questions — Topic 11.1

S1. Why is the left ventricular wall thicker than the right?

The left ventricle pumps blood to the entire body and must generate higher pressure, while the right ventricle pumps only to the lungs.

S2. How do heart valves ensure one-way blood flow?

Atrioventricular valves prevent backflow into atria during ventricular contraction, and semilunar valves prevent backflow into ventricles during relaxation.

S3. Why is there a delay at the AV node?

It allows atrial systole to complete and ventricles to fill before ventricular systole begins.

S4. How does the SA node act as a pacemaker?

It spontaneously generates electrical impulses that initiate each heartbeat and set the rhythm of contraction.

S5. What causes the "lubb-dubb" heart sounds?

"Lubb" is caused by closure of atrioventricular valves; "dubb" by closure of semilunar valves.

S6. Why may a heart murmur occur?

If valves do not close fully or open narrowly, turbulent blood flow produces a hissing sound called a murmur.

TOPIC 11.2 — BLOOD VESSELS

Textbook topic 11.2 · printed pages 238–246

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Q.3 Compare arteries, veins and capillaries. How is blood flow regulated in capillaries?

Board long question · 8 marks

Feature	Arteries	Veins	Capillaries
Direction	Carry blood away from heart	Carry blood toward heart	Connect arterioles and venules
Wall	Thick, muscular and elastic	Thinner, less muscular	Single layer of endothelial cells
Lumen	Narrow	Wide	Very narrow (~8 µm)
Valves	Absent	Present to prevent backflow	Absent
Pressure	High	Low	Low; allows exchange
Blood	Usually oxygenated (except pulmonary arteries)	Usually deoxygenated (except pulmonary veins)	Site of material exchange

Capillaries are the sites of exchange of materials between blood and tissues by diffusion, active transport, endocytosis and exocytosis. Fluid leaking from capillaries forms **interstitial fluid**.

Blood flow in capillaries is regulated by **pre-capillary sphincters** and by **vasoconstriction** and **vasodilation** of arterioles. Increased metabolic activity causes vasodilation, increasing blood flow to supply oxygen and remove CO_2.

Blood velocity is highest in the aorta (450–500 mm/s), lowest in capillaries (0.1 mm/s), and rises again in veins (250–300 mm/s), due to changes in total cross-sectional area.

Q.4 Describe pulmonary and systemic circulation.

Board long question · 8 marks

Pulmonary circulation

Deoxygenated blood leaves the right ventricle through the pulmonary trunk, which divides into pulmonary arteries carrying blood to the lungs. In lung capillaries, blood is oxygenated and returns through pulmonary veins to the left atrium.

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Systemic circulation

Oxygenated blood leaves the left ventricle through the aorta and supplies all body organs except the lungs. It returns deoxygenated blood to the right atrium through the venae cavae. It includes **coronary circulation** (supplying the heart wall), **hepatic portal circulation** (carrying absorbed nutrients to the liver) and **renal circulation** (supplying the kidneys).

Important Conceptual Short Questions — Topic 11.2

S1. Why are capillaries suited for exchange of materials?

Their walls are only one cell thick, their lumen is narrow, and they form extensive networks close to body cells, allowing efficient diffusion.

S2. How do pre-capillary sphincters regulate blood flow?

They contract or relax to control the amount of blood entering capillary loops, matching flow to tissue needs.

S3. Why does blood velocity decrease in capillaries?

The total cross-sectional area of the capillary network is very large, so blood flow slows down, allowing time for exchange.

S4. How does vasodilation help active tissues?

It increases blood flow, supplying more oxygen and nutrients and removing more carbon dioxide and wastes.

S5. Why do veins have valves while arteries do not?

Veins carry blood at low pressure back to the heart, so valves prevent backflow. Arteries carry blood at high pressure and do not need valves.

S6. How does coronary circulation support the heart?

It supplies oxygen and nutrients to the cardiac muscle through coronary arteries and removes wastes through coronary veins.

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TOPIC 11.3 — BLOOD PRESSURE

Textbook topic 11.3 · printed pages 246–249

Q.5 Explain blood pressure and its regulation.

Board long question · 8 marks

Blood pressure is the force exerted by blood against the inner walls of blood vessels. It is highest in the aorta and gradually reduces in systemic arteries. The rhythmic throbbing of arteries is called the **pulse**.

Systolic pressure is the maximum pressure during ventricular systole (normally about 120 mm Hg). **Diastolic pressure** is the lowest pressure during ventricular relaxation (conventionally about 80 mm Hg; the textbook also mentions 86 mm Hg in one line). Blood pressure is expressed as **120/80** and measured with a **sphygmomanometer**.

Short-term regulation

Baroreceptors in the carotid arteries and aortic arch detect changes in blood pressure. When pressure falls, they signal the brain to increase heart rate and force of contraction and cause vasoconstriction, restoring pressure.

Long-term regulation

- **ADH:** When blood volume and pressure fall, the posterior pituitary releases ADH, which makes kidneys retain water and constricts arterioles, raising blood pressure.
- **ANH:** When increased blood volume stretches the right atrium, it secretes atrial natriuretic hormone, which increases salt and water excretion, lowering blood volume and pressure.

Important Conceptual Short Questions — Topic 11.3

S1. Why is systolic pressure higher than diastolic pressure?

Systolic pressure occurs when ventricles contract and force blood into arteries, while diastolic pressure is the lower pressure remaining when ventricles relax.

S2. How do baroreceptors help maintain blood pressure?

They detect pressure changes and signal the brain to adjust heart rate, contraction force and arteriole diameter to restore normal pressure.

S3. How does ADH raise blood pressure?

It makes kidneys retain more water, increasing blood volume, and also constricts arterioles.

S4. Why does ANH lower blood pressure?

It is released when blood volume stretches the atrium and increases salt and water excretion, reducing blood volume and pressure.

S5. Why is blood pressure expressed as 120/80?

120 mm Hg is the normal systolic pressure during ventricular contraction, and 80 mm Hg is the normal diastolic pressure during relaxation.

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TOPIC 11.4 — CARDIOVASCULAR DISORDERS

Textbook topic 11.4 · printed pages 249–253

Q.6 Describe atherosclerosis, thrombosis, myocardial infarction and their treatments.

Board long question · 8 marks

- **Atherosclerosis:** Deposition of fatty materials, cholesterol, fibrin and cellular debris in arteries, narrowing them and increasing risk of thrombus and heart attack.
- **Arteriosclerosis:** Hardening of arterial walls due to calcium deposition, restricting blood flow and forcing the heart to work harder.
- **Thrombosis:** Formation of a thrombus (clot) that blocks a vessel. If dislodged, it becomes an **embolus**. Coronary thrombosis causes heart attack; cerebral thrombosis causes stroke.
- **Myocardial infarction:** Death of heart muscle due to blocked coronary blood supply. **Angina pectoris** is chest pain when oxygen demand exceeds supply.

Treatments

- **Angiography:** X-ray examination of coronary vessels using a contrast dye.
- **Coronary bypass surgery:** A healthy vessel is grafted to bypass the blocked coronary artery.
- **Angioplasty and stenting:** A balloon catheter dilates the narrowed artery, and a stent keeps it open.
- **Hypertension management:** Chronic BP above 140/90 increases risk of stroke, heart failure and kidney damage; lifestyle changes and medication are needed.

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Important Conceptual Short Questions — Topic 11.4

S1. How does atherosclerosis increase the risk of heart attack?

It narrows coronary arteries and promotes thrombus formation, reducing or blocking blood supply to heart muscle.

S2. Differentiate between thrombus and embolus.

A thrombus is a clot attached to a vessel wall. An embolus is a dislodged thrombus that travels through the bloodstream and may block another vessel.

S3. Why does angina pectoris occur?

It occurs when oxygen demand of heart muscle exceeds oxygen delivery, usually due to narrowed coronary arteries.

S4. How does coronary bypass surgery restore blood flow?

A healthy blood vessel is attached above and below the blocked coronary artery, allowing blood to bypass the obstruction.

S5. How does a stent help after angioplasty?

It is an expandable mesh tube that supports the artery wall and keeps the dilated artery open.

S6. Why is hypertension dangerous?

It increases the risk of blood vessel rupture, stroke, heart failure and kidney damage by forcing the heart and vessels to work under excessive pressure.

TOPIC 11.5 — LYMPHATIC SYSTEM OF HUMAN

Textbook topic 11.5 · printed pages 253–254

Q.7 Describe the lymphatic system and its functions.

Board long question · 8 marks

The lymphatic system consists of lymph vessels, lymphoid masses, lymph nodes and lymph. It begins with blind-ended **lymph capillaries** in interstitial fluid. These capillaries are more permeable than blood capillaries, so larger molecules can enter. When interstitial fluid enters lymph capillaries, it is called **lymph**.

Lymph capillaries join to form lymph vessels, which join into the **right lymphatic duct** and **thoracic duct**. These open into the right and left subclavian veins. Lymph flow is maintained by skeletal muscle activity, visceral movement and breathing, and valves prevent backflow.

Functions

- Returns excess fluid and dissolved proteins to the blood (about 3 litres daily).
- Absorbs and transports fats through **lacteals** in intestinal villi.
- Defends the body: lymph nodes filter lymph and contain lymphocytes and macrophages that destroy pathogens.
- The spleen filters blood, destroys aged RBCs and stores RBCs.

Important Conceptual Short Questions — Topic 11.5

S1. Why is the lymphatic system necessary alongside the blood circulatory system?

It returns excess interstitial fluid and proteins to the blood, transports fats and helps defend against infection.

S2. How do lacteals help in fat absorption?

They absorb chylomicrons from intestinal villi and transport them through lymph into the bloodstream.

S3. Why are lymph capillaries more permeable than blood capillaries?

They allow larger molecules such as proteins and chylomicrons to enter, which cannot easily enter blood capillaries.

S4. How do lymph nodes protect the body?

They filter lymph and contain lymphocytes and macrophages that destroy bacteria and viruses.

S5. What maintains the flow of lymph?

Skeletal muscle contractions, visceral movements, breathing movements and valves that prevent backflow.

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